

On the Origin of Stochastic, Low-Frequency Photometric Variability in Massive Stars

MATTEO CANTIELLO,^{1,2} DANIEL LECOANET,^{3,4,2} ADAM S. JERMYN,¹ AND LUCA GRASSITELLI⁵

¹*Center for Computational Astrophysics, Flatiron Institute, New York, NY 10010, USA*

²*Department of Astrophysical Sciences, Princeton University, Princeton, NJ 08544, USA*

³*Department of Engineering Sciences & Applied Mathematics, Northwestern University, Evanston, IL 60201, USA*

⁴*CIERA, Northwestern University, Evanston, IL 60201, USA*

⁵*Argelander-Institut für Astronomie der Universität Bonn, Auf dem Hügel 71, D-53121 Bonn, Germany*

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ABSTRACT

High-precision photometric observations have revealed ubiquitous stochastic low-frequency photometric variability in early type stars. It has been suggested that this variability arises due to either subsurface convection or internal gravity waves launched by the convective core. We show that an explanation for the observed surface photometric variability and macroturbulence relying on convective core driven internal gravity waves encounters a number of difficulties, and seems unlikely to be able to explain the observed trends. We suggest that low-frequency, stochastic photometric variability and surface turbulence in massive stars are caused by the the presence of subsurface convection.

Keywords: convection, stars: magnetic field, stars: variability

1. INTRODUCTION

Massive stars largely drive the dynamical and chemical evolution of gas in galaxies. They accomplish this via their stellar winds, eruptions, and explosive deaths. The legacy of massive stars is continued by the compact remnants they leave behind, neutron stars and black holes. These remnants can merge and generate the gravitational waves observed by LIGO/Virgo. The evolutionary trajectory starting with a massive star burning hydrogen in its core and ending with a compact remnant is understood only qualitatively. We still do not know how to map initial properties of the star, like mass, rotation rate, and metallicity, to e.g. the final mass and spin of the compact remnant it leaves behind. The picture is actually even more complicated, as the majority of massive stars is found in multiple systems (Sana et al. 2012), with a large fraction expected to interact with a companion during its lifetime (de Mink et al. 2014).

The path towards a detailed understanding of massive stars begins with a quantitative study of their internal structure on the main sequence. In recent years aster-

oseismology has opened a new window to the interior of these challenging astrophysical environments, with high precision photometry from space delivering many new exciting results (e.g., MOST, CoRoT, BRITE, Kepler/K2 and TESS, see Bowman 2020). The latest discovery is the detection of a new ubiquitous phenomenon in massive stars, stochastic low-frequency photometric variability (SLF variability; Bowman et al. 2019a,b; Pedersen et al. 2019; Bowman et al. 2020). This joins a number of other surface and wind phenomena that are routinely observed in early-type stars, including surface velocity fluctuations (Macroturbulence; Simón-Díaz & Herrero 2014), line profile variability (Fullerton et al. 1996), and discrete absorption components in UV spectra (Howarth & Prinja 1989; Cranmer & Owocki 1996; Fullerton et al. 1997; Kaper et al. 1997). Other phenomena that are harder to observe but could still be common in these intrinsically bright stars are surface magnetism and bright spots (e.g. Ramiamananantsoa et al. 2014). The origin of this SLF variability is currently debated. It could be caused by internal gravity waves (IGWs) launched by turbulent core convection (Edelmann et al. 2019; Ratnasingam et al. 2020), or by sub-surface convection zones (Cantiello et al. 2009; Blomme et al. 2011; Lecoanet et al. 2019). Regardless of its origin, this photometric signal is likely carrying important information

about the stellar structure, complementing asteroseismic studies that use well-identified oscillation modes (e.g. Aerts 2019; Aerts et al. 2019; Bowman 2020).

A recent development was the use of high resolution ground-based spectroscopy for the targets observed by K2 and TESS (Bowman et al. 2020). This allowed precise determination of stellar parameters, including mass, luminosity and macroturbulence. The mechanism exciting surface turbulent velocity fields is likely the same that produces the observed SLF variability (Bowman et al. 2020).

Here we used this data to compare the observed properties of the stochastic photometric variability and macroturbulence with some predictions coming from 1D stellar models. We discuss the properties of subsurface convection, in particular the expected trends in amplitude and frequency of the variability they could induce at the stellar surface, as a function of the stellar temperature and luminosity. We show that the predicted trends coming from a subsurface convection zone driven by the iron opacity peak at $\approx 150kK$ match very well the observations.

A connection between surface gas motion, line-broadening, and inefficient convection in stellar envelopes was also advocated by (Grassitelli et al. 2016), who used observations of hot massive stars in the Magellanic Clouds (Dufton et al. 2006). Moreover our result supports the suggestion of Penny & Gies (2009) that the processes leading to macroturbulence are likely metallicity dependent, since core convection is not expected to depend on the stellar metallicity but the presence and properties of the FeCZ do.

We show that a way to break the degeneracy between the two proposed mechanisms is looking at macroturbulence and SLF in massive stars with surface magnetic fields, since magnetic effects have a larger impact on core IGWs than on the FeCZ. Macroturbulence is observed in stars with fairly strong magnetic fields, sufficient to suppress IGWs from the core. On the other hands, the only stars consistent with having no macroturbulence are stars where the magnetic field is strong enough to shut off the FeCZ, consistent with a subsurface origin of surface perturbations.

We conclude suggesting that subsurface convection represents a possible unifying mechanism causing SLF variability, surface turbulence, and magnetic spots in massive stars.

2. METHODS

We calculated stellar evolution models using the Modules for Experiments in Stellar Astrophysics (MESA

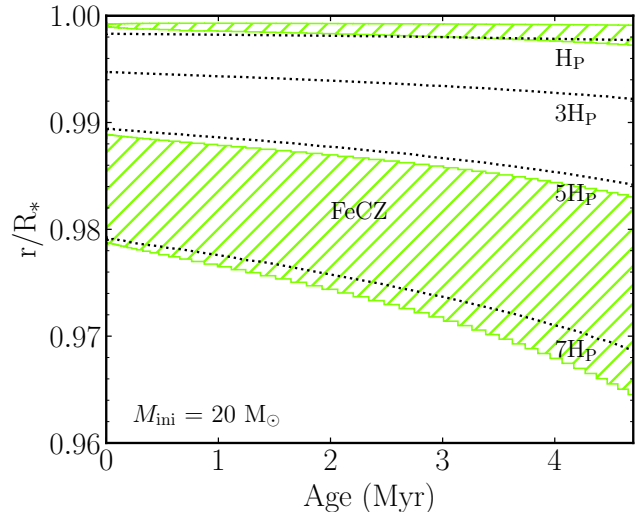


Figure 1. Evolution of the normalized radial location of the FeCZ in a $20M_{\odot}$ model during the main sequence (first 4.7 Myr of evolution). The HeCZ is also visible very close to the surface. The e -folding radial location of the local pressure respect to the surface pressure is shown as a dashed line marked by H_P . Locations of $P(r)/P(R_*) = e^n$ for $n = [3, 5, 7]$ are also shown.

Paxton et al. 2011, 2013, 2015, 2018, 2019) software instrument. Details on the microphysics inputs to this software instrument are given in Appendix A. We focused on the structure and convective properties of our models, in order to calculate typical frequencies and convective fluxes. Convection is calculated in the framework of the Mixing Length Theory (Böhm-Vitense 1958, MLT hereafter), and we adopted $\alpha_{MLT} = 1.6$. While in efficient convection zones (e.g. stellar cores) the choice of this parameter does not change results much, this is not true in superadiabatic, inefficient convective regions close to the stellar surface (Jiang et al. 2015; Cantiello & Braithwaite 2019). We discuss in Section 5 how uncertainties in the choice of the α_{MLT} parameter affect our results.

Since we are interested in the excitation of surface phenomena, we focus on the upper part of convective zones and following Cantiello et al. (2009) we define the average of a generic quantity q as

$$\bar{q} \equiv \frac{1}{\alpha_{MLT}H_P} \int_{R_c - \alpha_{MLT}H_P}^{R_c} q(r) dr \quad (1)$$

where H_P is the pressure scale height calculated at the upper boundary (R_c) of the convective zone of interest. We tested using different average procedures, and found that our results do not depend much on the specific choice.

Using this definition, we extracted the average Mach number $\bar{M}_c$, convective velocity $\bar{v}_c$ and density $\bar{\rho}_c$ in the convective core and in the subsurface convection zones of our models. The most important subsurface convection zone for the massive stars discussed in this work is the iron convection zone (FeCZ, see Fig. 1), although in the low luminosity regime He-driven convection zones could play a role as well (Cantiello et al. 2009; Cantiello & Braithwaite 2019). We define the characteristic frequency as

$$\nu_{\text{char}} \equiv \frac{1}{2\pi\tau}, \quad (2)$$

where τ is a characteristic timescale. When calculating the characteristic frequency in our models, we set τ to be the average convective turnover time

$$\tau_c = \alpha_{\text{MLT}} \bar{H}_P / \bar{v}_c \quad (3)$$

calculated either in the FeCZ or in the convective core. We proceeded to calculate convective turnover timescales and other relevant properties like convective fluxes $F_c = \bar{\rho}_c \bar{v}_c^3$. We did this as a function of both mass and evolutionary history for stars ranging from $5 - 120 M_\odot$. Here we present results for an initial metallicity of $Z=0.02$, but in Appendix C we report results for model grids with $Z = 0.014, 0.006$ and 0.002 as well.

We compare these quantities with the observed characteristic frequency ν_{char} and amplitude α_0 of SLF variability. These quantities are derived by fitting the stochastically-excited, low-frequency power excess in a power density spectrum using a Lorentzian function (e.g. Bowman et al. 2019b):

$$\alpha(\nu) = \frac{\alpha_0}{1 + \left(\frac{\nu}{\nu_{\text{char}}}\right)^\gamma} + P_W. \quad (4)$$

This shows that α_0 represents the amplitude at zero frequency. ν_{char} is defined as in Eq. 2 with τ the typical timescale of the SLF variability. Finally, γ is the gradient of the linear part of the profile and P_W is a white noise term. **Due to its stochastic, low-frequency manifestation in the power density spectrum, in the literature the SLF variability is also referred to as “red noise” (e.g. Blomme et al. 2011).**

3. RESULTS

We want to test a possible correlation between the properties of the FeCZ and observed photospheric phenomena, in particular SLF variability and turbulent velocities at the stellar surface (macroturbulence). We proceed with mapping quantities that measure the amplitude of perturbations at the location of the FeCZ. We then check if some of these properties correlate with the

amplitude of observed photospheric phenomena, including turbulent velocity fluctuations and SLF variability. While here we are agnostic about the particular mechanism, we assume that some physical process connects the subsurface layers to the photosphere. We discuss the most likely candidates for this process in Sec. 5.

3.1. FeCZ and Macroturbulence

We show the average convective velocity in the FeCZ in Fig. 2, left panel. Velocities of the order $10 \dots 100 \text{ km s}^{-1}$ are found across the upper H-R diagram, with a trend of increasing v_c for higher luminosities. Note that, contrary to core convection, the FeCZ is just mildly subsonic, with values of the Mach number ranging from 0.01 to 0.3. A caveat is the large uncertainty (a factor ≈ 8) of the convective velocities calculated using the MLT in inefficient convective zones (Cantiello & Braithwaite 2019). **In the right panel of the same figure we also show the maximum of the ratio of turbulent pressure to total pressure ($P_{\text{turb}} \propto v_c^2$), which, in agreement with the results of Grassitelli et al. (2015), shows a tight correlation with the spectroscopically-derived macroturbulent velocities in massive stars.** While they are both proxies for inefficient convection, the fraction of turbulent pressure and the average convective velocity over the uppermost pressure scale height of the convective zone slightly differ in the implied wave-generation mechanism. The first considers the waves as stochastically-generated by the Lagrangian perturbation of turbulent pressure, including finite-amplitude fluctuations following the local departure from hydrostatic equilibrium (Grassitelli et al. 2015). The latter invokes the interaction between the convective eddies and the overlying radiative layers (Cantiello et al. 2009), in which the convective elements conserve their inertia as they reach layers stable against convection, and transform part of their kinetic energy into stochastically-generated waves (e.g. Goldreich & Kumar 1990).

We confirm that convective velocities and the ratio of turbulent pressure to total pressure in the FeCZ correlate very well with the amplitude of macroturbulence.

3.2. FeCZ and Stochastic, Low-Frequency Variability

If the SLF variability was caused by the FeCZ, a natural choice of proxy for the typical timescale of this variability is the convective turnover timescale. We found typical values of the convective turnover timescale to be about $\sim 0.1 \dots 2 \text{ d}$ in the FeCZ, with a tendency for

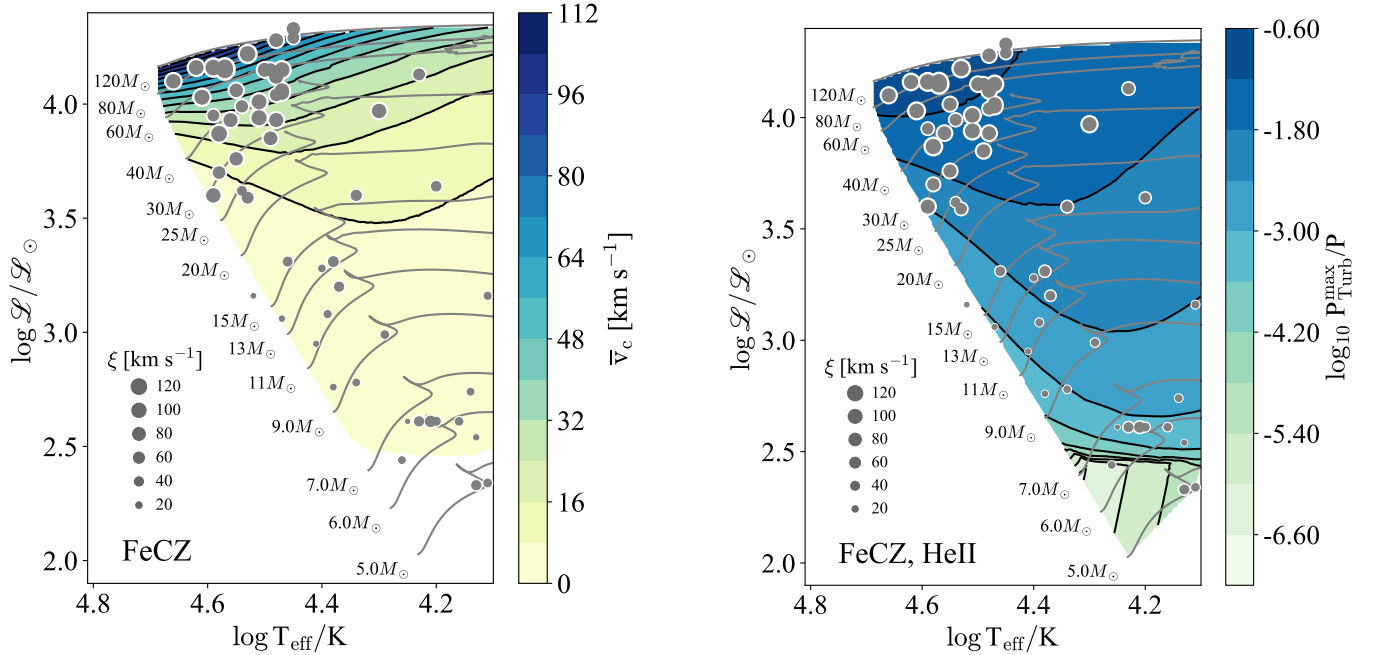


Figure 2. Left: Average convective velocities in the FeCZ as function of the location of stellar models in the spectroscopic H-R Diagram. We also show the observed stars with detected macroturbulent velocity as grey dots. The size of the dot is proportional to the observed macroturbulence. Right: Same as left, but showing predictions for the maximum of the ratio between turbulent pressure and total pressure in any subsurface convection zone. The FeCZ largely dominates, except for stars at low luminosity where turbulent pressure is provided by a helium convection zone.

shorter values in models with high effective temperature and surface luminosity (Cantiello et al. 2009). To quantitatively compare to the characteristic frequencies of the observed SLF variability in massive stars, we proceed mapping our theoretical predictions for the quantity ν_{char} (eq. 2) on a spectroscopic H-R diagram (Langer & Kudritzki 2014). This reveals a very good agreement, with our models reproducing both the typical values of the observed ν_{char} and the trend with $\log \mathcal{L}$ and $\log T_{\text{eff}}$ (Fig. 3). While the characteristic frequencies found by Bowman et al. (2020) seem to be larger by a factor of ≈ 3 , our predictions for the turnover timescale are affected by uncertainty in the convective velocities arising from the MLT treatment ($v_c \propto \alpha_{\text{MLT}}^3$, so $\nu_{\text{char}} \propto \alpha_{\text{MLT}}^2$). Since α_{MLT} is uncertain by a factor of 2 or so, our estimates of the characteristic frequency in the FeCZ are uncertain by a factor of ≈ 4 .

We also compare the amplitude of SLF variability with the relative convective flux in the FeCZ (left panel in Fig. 3). Values of F_c/F_* tend to increase with increasing $\log \mathcal{L}$ and decreasing $\log T_{\text{eff}}$, a trend that is also found for the values of observed SLF variability.

Overall the turnover timescale and relative flux in the FeCZ correlate very well with the timescale and amplitude of the observed SLF variability.

3.3. Core Convection and Stochastic, Low-Frequency Variability

Early-type stars possess convective cores during their main sequence. Turbulent convection in these convective cores excites internal gravity waves (Goldreich & Kumar 1990; Rogers et al. 2013; Lecoanet & Quataert 2013; Shiode et al. 2013; Edelmann et al. 2019; Horst et al. 2020), that can propagate through the stellar envelope and reach the stellar surface (e.g. Ratnasingam et al. 2019; Lecoanet et al. 2019; Ratnasingam et al. 2020). These waves could in principle be responsible for both the observed macroturbulence and SLF variability in early-type stars (Aerts et al. 2009; Simón-Díaz et al. 2017; Bowman et al. 2019b,a). Shiode et al. (2013) estimated that IGWs propagating from the core should lead to surface brightness fluctuations with typical amplitudes $\approx 10^{-2} - 10^2 \mu\text{mag}$, at odds with the relatively large amplitudes $\approx 10 - 10^4 \mu\text{mag}$ observed by e.g. Bowman et al. (2020). Lecoanet et al. (2019) also argue that the low-frequency, stochastic variability is unlikely to be caused by linearly propagating waves from the core. Their predicted wave transfer function shows little power at very low frequencies due to radiative damping, and standing modes at frequencies above 0.5 d^{-1} . None of these features are present in the observed spectra. Recent numerical simulations of wave generation by

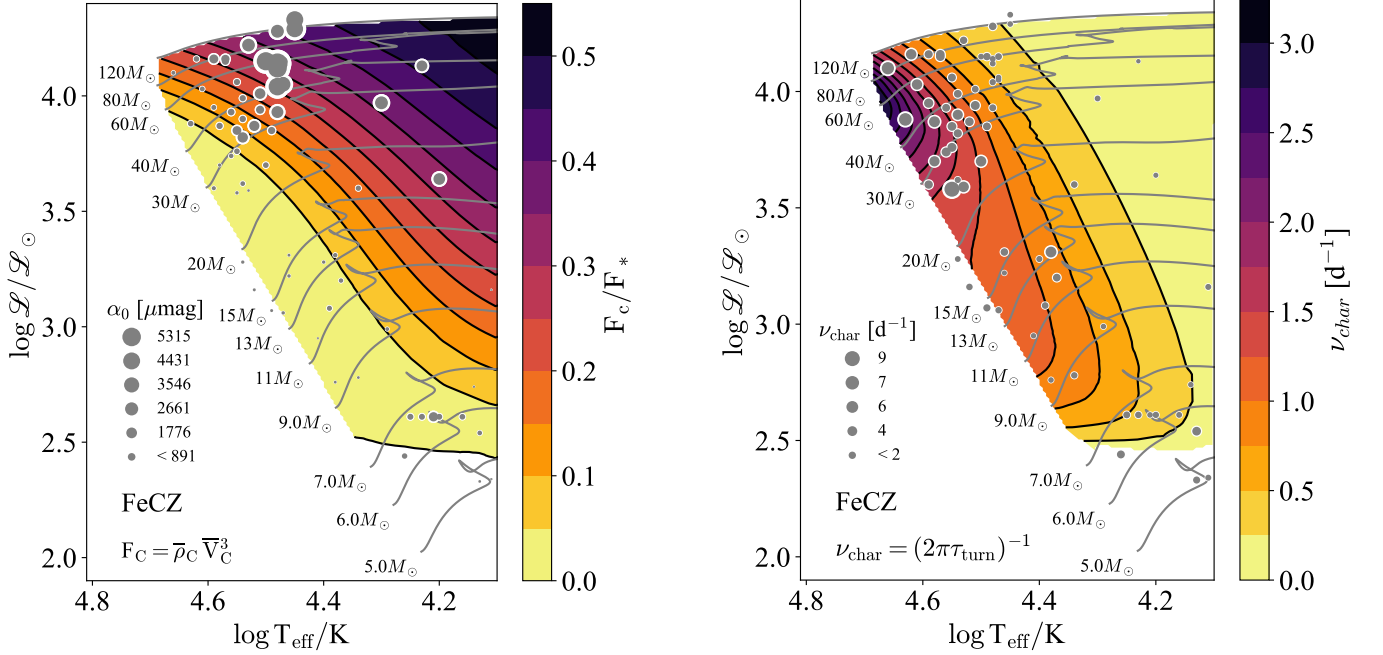


Figure 3. Left panel: Ratio of FeCZ convective flux to the total stellar flux in the spectroscopic H-R Diagram. We also show the observed stars with detected stochastic, low-frequency photometric variability as grey dots. The size of the dot is proportional to α_0 . Right panel: Characteristic frequency ν_{char} in the FeCZ as a function of the location of stellar models in the spectroscopic H-R Diagram (black contour lines). Evolutionary tracks are shown as grey solid lines. We also show the observed stars with detected stochastic photometric variability as grey dots. The size of the dot is proportional to ν_{char} .

convection in a $3M_\odot$ star (Edelmann et al. 2019; Horst et al. 2020) produce wave fluctuation spectra which are qualitatively similar to those observed. However, those simulations artificially boost the stellar luminosity by factors ranging from 10^3 to 10^7 . This makes it difficult to compare the amplitudes of the waves in those simulations to, e.g., Shiode et al. (2013), which predicts the surface variability in stars with realistic luminosities.

Although the physics of wave generation by convection is still uncertain, here we analyze the core convective properties of our models. If the surface variability is due to core convection, one would expect the characteristic frequency and amplitude of the SLF variability to correlate with the properties of the core convection. The flux of IGWs excited by turbulent convection at the core boundary is $F_{\text{IGW}} = F_* \bar{M}_c$ (Goldreich & Kumar 1990), where $\bar{M}_c$ is the average Mach number calculated using Eq. 1 and the local adiabatic sound speed. We expect the luminosity fluctuations to be proportional to the relative surface radial displacement ξ_r/R produced by these waves at the stellar surface (e.g. Dziembowski 1977; Aerts et al. 2010). It can be shown that $\xi_r/R \propto \sqrt{F_{\text{IGW}}/F_0}$ (see Appendix B). This quantity is presented in Fig. 4. One important caveat is that this estimate neglects the important role of radiative damping, which is essential in shaping both the

amplitude and the shape of the spectrum of waves at the surface (e.g. Lecoanet et al. 2019). The Mach number in the convective core is on the order of $10^{-3} \dots 10^{-4}$, with a trend of increasing values for lower $\log T_{\text{eff}}$ and higher $\log L$. Despite neglecting the important effect of radiative damping, the trend for the relative radial displacement at the surface do show some interesting correlations with the observed trend in SLF variability (left panel in Fig. 4).

Next, we focus on the characteristic timescale of waves excited by core convection. We expect the maximum of the IGW flux to be launched at frequencies close to $\nu_{\text{char}} = (2\pi\tau_c)^{-1}$, with τ_c defined in Eq. 3. Fig. 4 shows that the predicted values of ν_{char} are in the range $0.02 \dots 0.008 \text{ d}^{-1}$ ($0.2 \dots 0.08 \mu\text{Hz}$), in agreement with the results of Shiode et al. (2013). These values are about 2 orders of magnitude smaller than the typical characteristic frequencies of SLF variability observed by (Bowman et al. 2019b,a). As mentioned earlier, the caveat is that wave propagation through the stellar envelope affects the spectrum, changing the frequency of maximum power of waves reaching the surface (Lecoanet et al. 2019). Radiative damping efficiently suppresses low-frequency IGWs (because these waves have high radial wavenumbers), and the peak of the spectrum observed at the surface is expected to move to higher

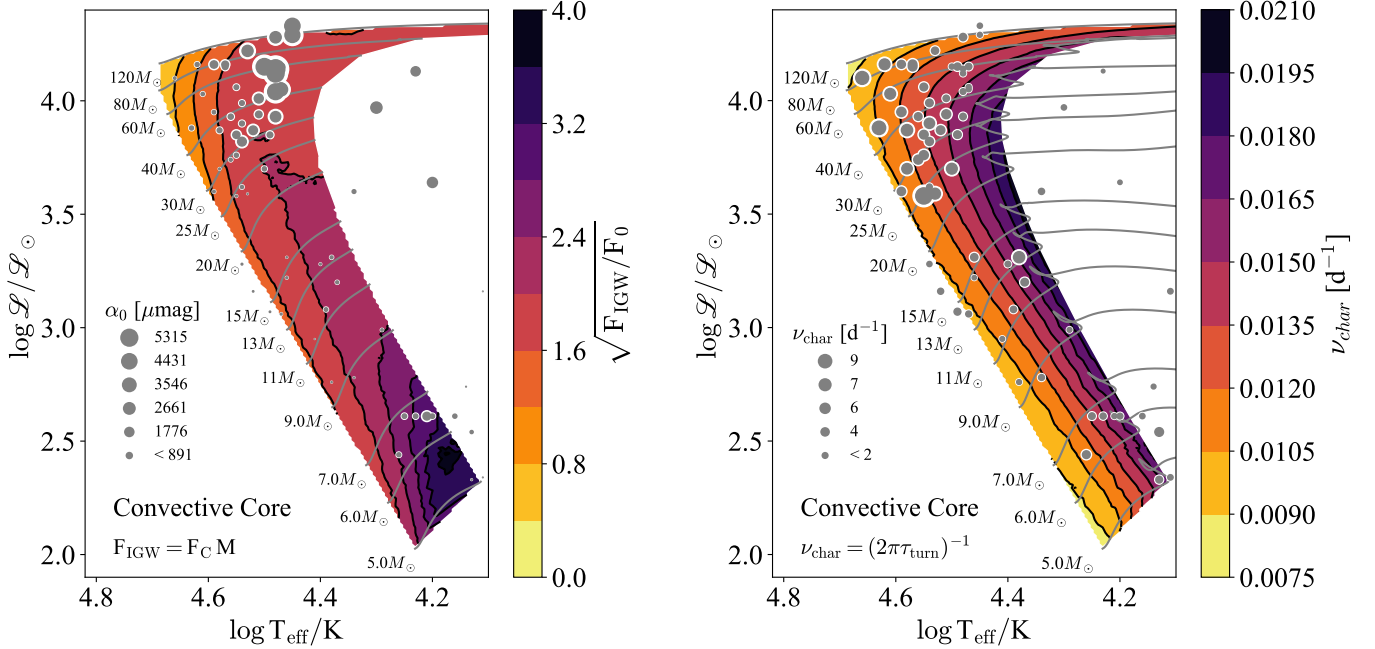


Figure 4. Left panel: Square root of the ratio between the flux of gravity waves launched by the convective core and $F_0 = 1/2\rho(r)r^3\omega^2\sqrt{N(r)^2 - \omega^2}$, evaluated at the stellar surface. In the absence of damping, this quantity is expected to be proportional to the relative radial displacement at the surface (See Appendix B). The IGW flux is calculated multiplying the core convective flux by the average convective Mach number in the top pressure scale height of the convective region. We also show the observed stars with detected stochastic, low-frequency photometric variability as grey dots. The size of the dot is proportional to α_0 . Right panel: Characteristic frequency ν_{char} in the convective core as function of the location of stellar models in the spectroscopic HRD. We also show the observed stars with detected photometric variability as grey dots. The size of the dot is proportional to ν_{char} .

frequencies. The amplitude of this effect depends on the envelope properties, which are a function of $\log T_{\text{eff}}$ and $\log L$. Nevertheless we will assume that trends of ν_{char} , as function of $\log T_{\text{eff}}$ and $\log L$, are still set by the core convective properties. Under this assumption, Fig. 4 shows that ν_{char} should increase with both $\log T_{\text{eff}}$ and $\log L$, so that the characteristic frequencies of peak IGWs should increase as stars evolve on the main sequence. This is exactly the opposite of what is observed: the characteristic frequencies of SLF variability is largest for stars early on on the main sequence, and seem to decrease as stars evolve. Therefore either radiative damping is able to revert this trend, or the observed variability might not be caused by IGWs launched by the core.

4. MACROTURBULENCE IN MAGNETIC STARS

The hypothesis that subsurface convection is responsible for surface turbulence is corroborated by the match between trends in the FeCZ properties (i.e. convective velocities and turbulent pressure) and the observed micro and macroturbulence (Cantiello et al. 2009; Grassitelli et al. 2015). One important test to this hypothesis is provided by the disappearance of macroturbulence in stars with surface magnetic fields above a critical

strength (Sundqvist et al. 2013), closely corresponding to the critical field needed to stabilize the FeCZ (MacDonald & Petit 2019; Jermyn & Cantiello 2020). Note that in OB stars, the FeCZ is deeper and more vigorous than the H and He convection zones, so a magnetic field stabilizing the FeCZ will necessarily also stabilize the other subsurface convection zones.

We can analogously define a critical magnetic field strength B_{crit} which suffices to reflect IGWs before they reach the photosphere. Using the dispersion relation for IGWs in a magnetic medium, the radial component of this field is (Fuller et al. 2015)

$$B_{r,\text{crit}} = \frac{\omega}{2k_r} \sqrt{4\pi\rho}, \quad (5)$$

where $\omega = \nu/2\pi$ is the angular frequency of the waves, ρ is the density, and k_r is the radial wave-number. For IGWs the radial wave-number is related to the azimuthal quantum number ℓ by

$$k_r \approx \frac{\sqrt{\ell(\ell+1)}}{r} \left(\frac{N}{\omega} \right), \quad (6)$$

where N is the Brünt-Väisälä frequency. So

$$B_{r,\text{crit}} \approx \frac{\omega^2 r}{N} \sqrt{\frac{4\pi\rho}{2\ell(\ell+1)}} \quad (7)$$

$$\approx \frac{\omega^2 r}{\ell N} \sqrt{4\pi\rho}. \quad (8)$$

Note that because this decreases with increasing ℓ , all waves of a given frequency are reflected if that with $\ell = 1$ is reflected.

We computed $B_{r,\text{crit}}$ for several values of ℓ and ν for a main-sequence model of a $30M_\odot$ star as a function of radius, shown in Fig. 5. For frequencies comparable to those of core convection the critical magnetic field strength is of order 10^{-2} G to 10^{-1} G. Strong macroturbulence is observed in similar-mass O-type stars with magnetic fields up to 2.5 kG (e.g. HD 191612, Sundqvist et al. 2013), so macroturbulence in those stars is unlikely to be due to IGWs coming from their cores if those waves have similar frequencies to that of core convection.

For frequencies comparable to the observed ν_{char} , on the order of 3 d^{-1} , the critical magnetic field is much larger, on the order of 300 G to 1 kG. This makes an explanation of macroturbulence based on IGWs marginally inconsistent with observations showing strong macroturbulence up to field strengths of 2.5 kG. However, because equation (7) is a strong function of ω , and hence of ν , it is possible that these strongly magnetized stars just have larger ν_{char} . For instance if HD 191612 has $\nu_{\text{char}} \approx 5\text{ d}^{-1}$ then it would have $B_{r,\text{crit}} \approx 3\text{ kG}$ for the $\ell = 1$ mode.

A further prediction of this calculation is that, if IGWs are the cause of macroturbulence in these stars, we should expect the strength of macroturbulence to decline with increasing magnetic field strength as more and more modes are reflected before they reach the surface. Sundqvist et al. (2013) find no such trend, and the only strong correlation they see of macroturbulence with magnetic field strength is that it vanishes when the magnetic field exceeds the FeCZ shutoff strength. This again points against an explanation of macroturbulence based on core-generated IGWs.

5. DISCUSSION

The presence of a subsurface convection zone can result in variability of photospheric properties via a number of processes, including wave excitation (Cantiello et al. 2009; Grassitelli et al. 2015) and magnetic buoyancy (Cantiello & Braithwaite 2011). Multi dimensional simulations including radiation have been performed in a restricted range of the parameter space, and show that the full turbulent manifestation of these convective regions extends up to the stellar surface (Jiang et al. 2015,

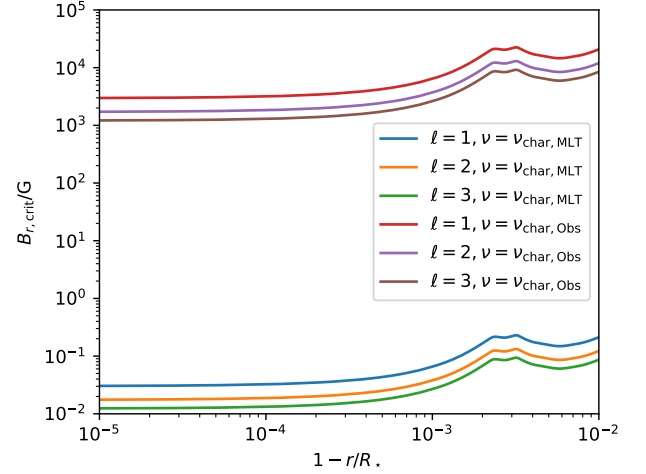


Figure 5. The critical radial magnetic field strength needed to reflect internal gravity waves of different ℓ and ν is shown as a function of fractional depth for a $30M_\odot$ stellar model at an age of 4 Myr. The frequency $\nu_{\text{char,MLT}}$ is that given by equation (2) but evaluated for the core convection zone instead of the subsurface FeCZ. The frequency $\nu_{\text{char,Obs}}$ is a typical value of 3 d^{-1} for the observations.

2017, 2018). In their calculations of OB stars envelopes including the stellar photosphere, Jiang et al. (2015) observe turbulent velocities reaching the isothermal sound speed ($\approx 50\text{ km s}^{-1}$) at the stellar surface, demonstrating that velocity fields of amplitude comparable to the observed macroturbulence are naturally explained by the presence of the FeCZ.

Therefore it could be that the observed SLF variability and macroturbulence simply represent the direct manifestation of turbulent, radiation-dominated convection at the stellar surface. The simplified MLT treatment in our one dimensional calculations is unable to capture the complex phenomenology of these layers, but the fact that it can reproduce both the timescales and the trends in amplitude of the observed SLF variability is compelling. It calls for extending the radiation hydrodynamics simulations to cover the parameter space of the TESS observations, in order to unravel the precise mechanism connecting subsurface convection zones to the observed surface variability.

If the FeCZ is responsible for surface macroturbulence and SLF variability, then these phenomena should be affected by the stellar metallicity. On the other hand we do not expect a significant metallicity dependence in the context of a core-convection origin. The results of (Bowman et al. 2019b) show that SLF variability is also observed in low-metallicity, LMC stars. The presence of the FeCZ

depends on the luminosity and metallicity of the star, and in 1D stellar evolution calculations the FeCZ occurs above a luminosity of $L \approx 10^{3.2} L_{\odot}$ for $Z=0.02$. This corresponds to a zero age main sequence star of about $7M_{\odot}$ (see e.g. Fig. 2). The LMC has a metallicity about half solar, and in models at $Z=0.008$ the FeCZ only appears at $L \approx 10^{3.9} L_{\odot}$, corresponding to a zero age main sequence star of about $11M_{\odot}$ (Cantiello et al. 2009). Note that these limits could move downward in the presence of atomic diffusion and radiative acceleration (Richer et al. 2000), or due to an upward revision of the uncertain values of Fe opacity (e.g. Bailey et al. 2015). It will be interesting to see if the TESS observations can probe the transition region between stars with and without a FeCZ, and determine a possible change in surface properties. **At the same time, X-shooter within the Ulysses program could more firmly establish the metallicity-dependence of the macroturbulent line-broadening.**

It is important to note that some low-luminosity, main sequence A stars in Bowman et al. (2019b) show SLF variability. For these stars models do not predict the presence of the FeCZ. However, these stars still show (sub)surface convection zones triggered by ionization of H and He (Cantiello & Braithwaite 2019). The characteristic frequencies of convection in these regions are also in the range $\sim$ tens of μHz , although the amplitude of the velocity fluctuations they can induce is much smaller than for the FeCZ, so it is not clear if they could be linked to observed SLF variability and macroturbulence. On the other hand, since the relative perturbation induced by subsurface convection is weakest in the regime of A and late-B type (Cantiello & Braithwaite 2019; Jermyn & Cantiello 2020), these stars are the best targets for detecting core-generated IGWs. Depending on the amplitude of core generated IGWs, the impact of subsurface convection could well be subdominant in these objects, allowing for a detection. Shiode et al. (2013) predict that in this regime, the peak rms surface brightness fluctuations induced by core-generated IGWs should be in the range $\approx 0.01\text{-}100\mu\text{mag}$ (See their Fig. 6).

5.1. Stochastic, Low-Frequency Variability in Evolved Stars

SLF variability with similar properties to the massive main sequence stars discussed by (Bowman et al. 2019b, 2020) was recently observed in evolved massive stars. Naze et al. (2021) detected SLF variability in both luminous blue variables (LBV) and Wolf-Rayet (WR) stars, and Dorn-Wallenstein et al. (2020) found the same photometric signature in yellow supergiants (YSG). Com-

pared to OB stars, LBV, WR and YSG correspond to later stages of evolution. In particular, WR stars and YSG are likely burning helium in their cores. The internal structure of OB, LBV, WR and YSG stars changes dramatically, and this can affect substantially the generation and propagation of internal gravity waves. In particular, the amount of radiative damping is expected to change due to the sheer differences in envelope temperature and densities. We recall here that the radiative damping rate γ_{rad} for a traveling g-mode is given by

$$\gamma_{\text{rad}}(\omega, \ell, r) = K_{\text{rad}}(r) k_r^2,$$

$$K_{\text{rad}}(r) = \frac{16 \sigma T(r)^3}{3 \rho(r)^2 \kappa(r) c_p(r)}.$$

where k_r is the radial wavenumber (eqn. 6) and κ is the opacity. Compact WR stars have surface temperatures that can exceed $\approx 10^5\text{K}$, while OB stars and LBVs have effective temperatures $\approx 10^{4.7\text{--}4.4}\text{K}$, with LBVs found at the cooler end of this range. The surface of extended, low-density, YSGs is cooler than 10^4K . With different types of core convection (H-burning vs He-burning) and radiative damping rates, it would be surprising if core-generated IGWs in e.g. OB and WR stars showed up at the surface with similar properties. On the other hand, the FeCZ is present below/at the surface in both OB and WR stars, and with similar velocities and convective turnover times. This seems to strengthen the main thesis of this work, adding support to a (sub)surface origin of the observed SLF variability.

The discussion is a bit more complicated for LBV and YSG, since at lower temperatures other convective regions can become prominent (driven by H and He recombination, see e.g. Jiang et al. 2018). Dorn-Wallenstein et al. (2020) disfavor a near-surface origin for the observed SLF variability in YSG, on the ground that the observed timescale do not follow the predicted scaling for granulation (Kallinger et al. 2014). However, we believe that the scaling of Kallinger et al. (2014) is not applicable in the regime explored by Dorn-Wallenstein et al. (2020). This scaling was derived for a sample of red giants and solar-like stars, which all have temperatures well-below the temperature for the recombination of hydrogen, and it is not directly applicable to earlier spectral subtypes. Interestingly, the characteristic timescales of 0.1-1 days found by Dorn-Wallenstein et al. (2020) at $\log T_{\text{eff}} \approx 4$ are consistent with the convective turnover timescale in the FeCZ (e.g. Fig. 3). Moreover, in their lower temperature sample ($\log T_{\text{eff}} < 3.75$) the rapid increase in amplitudes and characteristic frequen-

cies of the variability is in agreement with the development of near-surface convection induced by the large opacities associated with the recombination of hydrogen. We suggest that surface and near-surface convection could be indeed responsible for the variability observed by [Dorn-Wallenstein et al. \(2020\)](#), and we plan to systematically study the properties of (sub)surface convection in these evolved, cool stars in future work.

5.2. *Towards a unified model for surface phenomena in massive stars*

The presence of subsurface convection can simultaneously account for a large variety of puzzling phenomena that appear ubiquitous at the surface of early-type stars. First we have microturbulence and macroturbulence, and both can be accounted for by velocity fields excited by the underlying FeCZ subsurface convection zone ([Cantiello et al. 2009](#); [Grassitelli et al. 2015](#); [Jiang et al. 2015](#)). The only stars with no macroturbulence appear to have magnetic fields strong enough to shut off convection in the FeCZ, while stars with slightly lower surface magnetic fields show normal values of macroturbulence ([Sundqvist et al. 2013](#)). Then we have the observation of bright spots in early-type stars (e.g. [Ramiaramanantsoa et al. 2014](#)), which can be explained with the presence of magnetic spots rising from subsurface convective layers ([Cantiello & Braithwaite 2011, 2019](#)). These photospheric magnetic spots can then be responsible for another ubiquitous phenomena in early-type stars, the presence of discrete absorption components (DACs) in their UV spectra (e.g. [Howarth & Prinja 1989](#); [Cranmer & Owocki 1996](#); [Fullerton et al. 1997](#); [Kaper et al. 1997](#)). Overall subsurface convection is expected to drive velocity and density perturbations at the stellar surface (e.g. [Cantiello et al. 2009](#); [Jiang et al. 2015](#)). These perturbations could explain the ubiquitous line profile variability in hot stars ([Fullerton et al. 1996](#)) and provide a seed for the rapid development of wind instabilities that lead to wind clumping ([Owocki et al. 1988](#)), but see also ([Sundqvist & Owocki 2013](#)). Finally, these surface velocity and density perturbations would necessarily induce photometric variability, and we have shown here that the observed SLF variability detected in OB stars correlates very well with the presence and properties of the FeCZ.

An economical hypothesis emerges: the presence of subsurface convection, and in particular of the FeCZ, could be the common underlying physical cause for the appearance of turbulence, magnetic spots, SLF variability, as well spectroscopic variability in early-type stars.

6. CONCLUSIONS

We used one-dimensional stellar evolution calculations to study the predicted trends in the properties of subsurface convection in the spectroscopic HR-Diagram. We found that the trends of relative convective flux and convective turnover timescale in the FeCZ of our models match very well the trends in timescale and amplitude of stochastic, low-frequency photometric variability in OB stars observed by TESS and K2.

Similar to previous works, we show that the observed trends in stellar macroturbulence are also well reproduced assuming the FeCZ as its driver. This connection is also supported by the observations of strongly magnetized early-type stars, showing no macroturbulence only for magnetic fields above the critical value required to shut-off turbulent convection in the FeCZ. We find that IGWs coming from the stellar core would be reflected or damped for values of the magnetic field well below this critical value. The fact that stars with strong but subcritical magnetic fields show typical values of macroturbulence points against a convective core origin of this surface perturbation.

Overall the observations support a picture in which subsurface convection, and in particular the FeCZ, is responsible for the ubiquitous low-frequency, stochastic photometric variability and macroturbulence detected in OB stars. These surface manifestations join a number of phenomena observed in early-type stars and attributed to the presence of subsurface convection, including the observations of (magnetic) bright spots as well as wind and spectroscopic variability. Radiation (magneto)hydrodynamics simulations of the outer envelope regions of early-type stars are required to understand the details of how subsurface convection zones cause the observed surface perturbations.

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APPENDIX

A. SOFTWARE DETAILS

Calculations were done with **MESA** version 11701. The **MESA** EOS is a blend of the OPAL Rogers & Nayfonov (2002), SCVH Saumon et al. (1995), PTEH Pols et al. (1995), HELM Timmes & Swesty (2000), and PC Potekhin & Chabrier (2010) EOSes. Radiative opacities are primarily from OPAL (Iglesias & Rogers 1993, 1996), with low-temperature data from Ferguson et al. (2005) and the high-temperature, Compton-scattering dominated regime by Buchler & Yueh (1976). Electron conduction opacities are from Cassisi et al. (2007). Nuclear reaction rates are a combination of rates from NACRE (Angulo et al. 1999), JINA REACLIB (Cyburt et al. 2010), plus additional tabulated weak reaction rates Fuller et al. (1985); Oda et al. (1994); Langanke & Martínez-Pinedo (2000). Screening is included via the prescription of Chugunov et al. (2007). Thermal neutrino loss rates are from Itoh et al. (1996). The inlists, processing scripts, and model output are available at Zenodo.org.

B. RADIAL DISPLACEMENT FROM IGWS

Let's consider traveling internal gravity waves in the WKB approximation. The wave flux is $F_{\text{IGW}} = \frac{1}{2}\rho|u|^2u_{g,r}$ with the group velocity $u_{g,r} = \partial\omega/\partial\kappa$. Since $u_{g,r} = (\omega/N)^2\sqrt{N^2 - \omega^2}/k_h$, $|u|^2 = (N/\omega)^2u_r^2$, and $u_r = \omega\xi_r$ we obtain:

$$F_{\text{IGW}} = \frac{1}{2}\rho\xi_r^2\omega^2\frac{\sqrt{N^2 - \omega^2}}{k_h}. \quad (\text{B1})$$

Using $k_h = \sqrt{\ell(\ell+1)}/r$ we can then write the relative radial displacement as:

$$\frac{\xi_r}{r} = \left[\frac{F_{\text{IGW}}\sqrt{\ell(\ell+1)}}{1/2\rho r^3\omega^2\sqrt{N^2 - \omega^2}} \right]^{1/2} \propto \sqrt{\frac{F_{\text{IGW}}}{F_0}} \quad (\text{B2})$$

C. GRIDS AT DIFFERENT METALLICITIES

Here we present results for the critical and surface magnetic field at metallicities of $Z=0.014$, $Z=0.006$, and $Z=0.002$, representing early-type stars in the Galaxy (MW), Large Magellanic Cloud (LMC), and Small Magellanic Cloud (SMC) respectively (e.g. Yusof et al. 2013). The initial Helium content of the grids is $Y=0.2659$ (MW), $Y=0.2559$ (LMC), and $Y=0.2508$ (SMC). The metallicity is initialized scaling the standard solar composition of Grevesse & Sauval (1998).

REFERENCES

- Aerts, C. 2019, arXiv e-prints, arXiv:1912.12300.
<https://arxiv.org/abs/1912.12300>
- Aerts, C., Christensen-Dalsgaard, J., & Kurtz, D. W. 2010, *Asteroseismology*
- Aerts, C., Mathis, S., & Rogers, T. M. 2019, *ARA&A*, 57, 35, doi: [10.1146/annurev-astro-091918-104359](https://doi.org/10.1146/annurev-astro-091918-104359)
- Aerts, C., Puls, J., Godart, M., & Dupret, M. A. 2009, *A&A*, 508, 409, doi: [10.1051/0004-6361/200810471](https://doi.org/10.1051/0004-6361/200810471)
- Angulo, C., Arnould, M., Rayet, M., et al. 1999, *Nuclear Physics A*, 656, 3, doi: [10.1016/S0375-9474\(99\)00030-5](https://doi.org/10.1016/S0375-9474(99)00030-5)
- Bailey, J. E., Nagayama, T., Loisel, G. P., et al. 2015, *Nature*, 517, 56, doi: [10.1038/nature14048](https://doi.org/10.1038/nature14048)
- Blomme, R., Mahy, L., Catala, C., et al. 2011, *A&A*, 533, A4, doi: [10.1051/0004-6361/201116949](https://doi.org/10.1051/0004-6361/201116949)
- Böhm-Vitense, E. 1958, *ZA*, 46, 108
- Bowman, D. M. 2020, arXiv e-prints, arXiv:2008.11162.
<https://arxiv.org/abs/2008.11162>
- Bowman, D. M., Burssens, S., Simón-Díaz, S., et al. 2020, *A&A*, 640, A36, doi: [10.1051/0004-6361/202038224](https://doi.org/10.1051/0004-6361/202038224)
- Bowman, D. M., Burssens, S., Pedersen, M. G., et al. 2019a, *Nature Astronomy*, 3, 760, doi: [10.1038/s41550-019-0768-1](https://doi.org/10.1038/s41550-019-0768-1)
- Bowman, D. M., Aerts, C., Johnston, C., et al. 2019b, *A&A*, 621, A135, doi: [10.1051/0004-6361/201833662](https://doi.org/10.1051/0004-6361/201833662)
- Buchler, J. R., & Yueh, W. R. 1976, *ApJ*, 210, 440, doi: [10.1086/154847](https://doi.org/10.1086/154847)
- Cantiello, M., & Braithwaite, J. 2011, *A&A*, 534, A140, doi: [10.1051/0004-6361/201117512](https://doi.org/10.1051/0004-6361/201117512)
- . 2019, *ApJ*, 883, 106, doi: [10.3847/1538-4357/ab3924](https://doi.org/10.3847/1538-4357/ab3924)
- Cantiello, M., Langer, N., Brott, I., et al. 2009, *A&A*, 499, 279, doi: [10.1051/0004-6361/200911643](https://doi.org/10.1051/0004-6361/200911643)
- Cassisi, S., Potekhin, A. Y., Pietrinferni, A., Catelan, M., & Salaris, M. 2007, *ApJ*, 661, 1094, doi: [10.1086/516819](https://doi.org/10.1086/516819)
- Chugunov, A. I., Dewitt, H. E., & Yakovlev, D. G. 2007, *PhRvD*, 76, 025028, doi: [10.1103/PhysRevD.76.025028](https://doi.org/10.1103/PhysRevD.76.025028)
- Cranmer, S. R., & Owocki, S. P. 1996, *ApJ*, 462, 469, doi: [10.1086/177166](https://doi.org/10.1086/177166)
- Cyburt, R. H., Amthor, A. M., Ferguson, R., et al. 2010, *ApJS*, 189, 240, doi: [10.1088/0067-0049/189/1/240](https://doi.org/10.1088/0067-0049/189/1/240)

- de Mink, S. E., Sana, H., Langer, N., Izzard, R. G., & Schneider, F. R. N. 2014, *ApJ*, 782, 7, doi: [10.1088/0004-637X/782/1/7](https://doi.org/10.1088/0004-637X/782/1/7)
- Dorn-Wallenstein, T. Z., Levesque, E. M., Neugent, K. F., et al. 2020, *ApJ*, 902, 24, doi: [10.3847/1538-4357/abb318](https://doi.org/10.3847/1538-4357/abb318)
- Dufton, P. L., Ryans, R. S. I., Simón-Díaz, S., Trundle, C., & Lennon, D. J. 2006, *A&A*, 451, 603, doi: [10.1051/0004-6361:20054600](https://doi.org/10.1051/0004-6361:20054600)
- Dziembowski, W. 1977, *AcA*, 27, 203
- Edelmann, P. V. F., Ratnasingam, R. P., Pedersen, M. G., et al. 2019, *ApJ*, 876, 4, doi: [10.3847/1538-4357/ab12df](https://doi.org/10.3847/1538-4357/ab12df)
- Ferguson, J. W., Alexander, D. R., Allard, F., et al. 2005, *ApJ*, 623, 585, doi: [10.1086/428642](https://doi.org/10.1086/428642)
- Fuller, G. M., Fowler, W. A., & Newman, M. J. 1985, *ApJ*, 293, 1, doi: [10.1086/163208](https://doi.org/10.1086/163208)
- Fuller, J., Cantiello, M., Stello, D., Garcia, R. A., & Bildsten, L. 2015, *Science*, 350, 423, doi: [10.1126/science.aac6933](https://doi.org/10.1126/science.aac6933)
- Fullerton, A. W., Gies, D. R., & Bolton, C. T. 1996, *ApJS*, 103, 475, doi: [10.1086/192285](https://doi.org/10.1086/192285)
- Fullerton, A. W., Massa, D. L., Prinja, R. K., Owocki, S. P., & Cranmer, S. R. 1997, *A&A*, 327, 699
- Goldreich, P., & Kumar, P. 1990, *ApJ*, 363, 694, doi: [10.1086/169376](https://doi.org/10.1086/169376)
- Grassitelli, L., Fossati, L., Langer, N., et al. 2016, *A&A*, 593, A14, doi: [10.1051/0004-6361/201628912](https://doi.org/10.1051/0004-6361/201628912)
- Grassitelli, L., Fossati, L., Simón-Díaz, S., et al. 2015, *ApJL*, 808, L31, doi: [10.1088/2041-8205/808/1/L31](https://doi.org/10.1088/2041-8205/808/1/L31)
- Grevesse, N., & Sauval, A. J. 1998, *SSRv*, 85, 161, doi: [10.1023/A:1005161325181](https://doi.org/10.1023/A:1005161325181)
- Horst, L., Edelmann, P. V. F., Andr  ssy, R., et al. 2020, *A&A*, 641, A18, doi: [10.1051/0004-6361/202037531](https://doi.org/10.1051/0004-6361/202037531)
- Howarth, I. D., & Prinja, R. K. 1989, *ApJS*, 69, 527, doi: [10.1086/191321](https://doi.org/10.1086/191321)
- Iglesias, C. A., & Rogers, F. J. 1993, *ApJ*, 412, 752, doi: [10.1086/172958](https://doi.org/10.1086/172958)
- . 1996, *ApJ*, 464, 943, doi: [10.1086/177381](https://doi.org/10.1086/177381)
- Itoh, N., Hayashi, H., Nishikawa, A., & Kohyama, Y. 1996, *ApJS*, 102, 411, doi: [10.1086/192264](https://doi.org/10.1086/192264)
- Jermyn, A. S., & Cantiello, M. 2020, *ApJ*, 900, 113, doi: [10.3847/1538-4357/ab9e70](https://doi.org/10.3847/1538-4357/ab9e70)
- Jiang, Y.-F., Cantiello, M., Bildsten, L., Quataert, E., & Blaes, O. 2015, *ApJ*, 813, 74, doi: [10.1088/0004-637X/813/1/74](https://doi.org/10.1088/0004-637X/813/1/74)
- . 2017, *ApJ*, 843, 68, doi: [10.3847/1538-4357/aa77b0](https://doi.org/10.3847/1538-4357/aa77b0)
- Jiang, Y.-F., Cantiello, M., Bildsten, L., et al. 2018, *Nature*, 561, 498, doi: [10.1038/s41586-018-0525-0](https://doi.org/10.1038/s41586-018-0525-0)
- Kallinger, T., De Ridder, J., Hekker, S., et al. 2014, *A&A*, 570, A41, doi: [10.1051/0004-6361/201424313](https://doi.org/10.1051/0004-6361/201424313)
- Kaper, L., Henrichs, H. F., Fullerton, A. W., et al. 1997, *A&A*, 327, 281
- Langanke, K., & Mart  nez-Pinedo, G. 2000, *Nuclear Physics A*, 673, 481, doi: [10.1016/S0375-9474\(00\)00131-7](https://doi.org/10.1016/S0375-9474(00)00131-7)
- Langer, N., & Kudritzki, R. P. 2014, *A&A*, 564, A52, doi: [10.1051/0004-6361/201423374](https://doi.org/10.1051/0004-6361/201423374)
- Lecoanet, D., & Quataert, E. 2013, *MNRAS*, 430, 2363, doi: [10.1093/mnras/stt055](https://doi.org/10.1093/mnras/stt055)
- Lecoanet, D., Cantiello, M., Quataert, E., et al. 2019, *ApJL*, 886, L15, doi: [10.3847/2041-8213/ab5446](https://doi.org/10.3847/2041-8213/ab5446)
- MacDonald, J., & Petit, V. 2019, *MNRAS*, 487, 3904, doi: [10.1093/mnras/stz1545](https://doi.org/10.1093/mnras/stz1545)
- Naze, Y., Rauw, G., & Gosset, E. 2021, arXiv e-prints, arXiv:2101.05521. <https://arxiv.org/abs/2101.05521>
- Oda, T., Hino, M., Muto, K., Takahara, M., & Sato, K. 1994, *Atomic Data and Nuclear Data Tables*, 56, 231, doi: [10.1006/adnd.1994.1007](https://doi.org/10.1006/adnd.1994.1007)
- Owocki, S. P., Castor, J. I., & Rybicki, G. B. 1988, *ApJ*, 335, 914, doi: [10.1086/166977](https://doi.org/10.1086/166977)
- Paxton, B., Bildsten, L., Dotter, A., et al. 2011, *ApJS*, 192, 3, doi: [10.1088/0067-0049/192/1/3](https://doi.org/10.1088/0067-0049/192/1/3)
- Paxton, B., Cantiello, M., Arras, P., et al. 2013, *ApJS*, 208, 4, doi: [10.1088/0067-0049/208/1/4](https://doi.org/10.1088/0067-0049/208/1/4)
- Paxton, B., Marchant, P., Schwab, J., et al. 2015, *ApJS*, 220, 15, doi: [10.1088/0067-0049/220/1/15](https://doi.org/10.1088/0067-0049/220/1/15)
- Paxton, B., Schwab, J., Bauer, E. B., et al. 2018, *ApJS*, 234, 34, doi: [10.3847/1538-4365/aaa5a8](https://doi.org/10.3847/1538-4365/aaa5a8)
- Paxton, B., Smolec, R., Schwab, J., et al. 2019, *ApJS*, 243, 10, doi: [10.3847/1538-4365/ab2241](https://doi.org/10.3847/1538-4365/ab2241)
- Pedersen, M. G., Chowdhury, S., Johnston, C., et al. 2019, *ApJL*, 872, L9, doi: [10.3847/2041-8213/ab01e1](https://doi.org/10.3847/2041-8213/ab01e1)
- Penny, L. R., & Gies, D. R. 2009, *ApJ*, 700, 844, doi: [10.1088/0004-637X/700/1/844](https://doi.org/10.1088/0004-637X/700/1/844)
- Pol, O. R., Tout, C. A., Eggleton, P. P., & Han, Z. 1995, *MNRAS*, 274, 964, doi: [10.1093/mnras/274.3.964](https://doi.org/10.1093/mnras/274.3.964)
- Potekhin, A. Y., & Chabrier, G. 2010, *Contributions to Plasma Physics*, 50, 82, doi: [10.1002/ctpp.201010017](https://doi.org/10.1002/ctpp.201010017)
- Ramaramananantsoa, T., Moffat, A. F. J., Chen  , A.-N., et al. 2014, *MNRAS*, 441, 910, doi: [10.1093/mnras/stu619](https://doi.org/10.1093/mnras/stu619)
- Ratnasingam, R. P., Edelmann, P. V. F., & Rogers, T. M. 2019, *MNRAS*, 482, 5500, doi: [10.1093/mnras/sty3086](https://doi.org/10.1093/mnras/sty3086)
- . 2020, *MNRAS*, doi: [10.1093/mnras/staa2296](https://doi.org/10.1093/mnras/staa2296)
- Richer, J., Michaud, G., & Turcotte, S. 2000, *ApJ*, 529, 338, doi: [10.1086/308274](https://doi.org/10.1086/308274)
- Rogers, F. J., & Nayfonov, A. 2002, *ApJ*, 576, 1064, doi: [10.1086/341894](https://doi.org/10.1086/341894)
- Rogers, T. M., Lin, D. N. C., McElwaine, J. N., & Lau, H. H. B. 2013, *ApJ*, 772, 21, doi: [10.1088/0004-637X/772/1/21](https://doi.org/10.1088/0004-637X/772/1/21)

- Sana, H., de Mink, S. E., de Koter, A., et al. 2012, *Science*, 337, 444, doi: [10.1126/science.1223344](https://doi.org/10.1126/science.1223344)
- Saumon, D., Chabrier, G., & van Horn, H. M. 1995, *ApJS*, 99, 713, doi: [10.1086/192204](https://doi.org/10.1086/192204)
- Shiode, J. H., Quataert, E., Cantiello, M., & Bildsten, L. 2013, *MNRAS*, 430, 1736, doi: [10.1093/mnras/sts719](https://doi.org/10.1093/mnras/sts719)
- Simón-Díaz, S., Godart, M., Castro, N., et al. 2017, *A&A*, 597, A22, doi: [10.1051/0004-6361/201628541](https://doi.org/10.1051/0004-6361/201628541)
- Simón-Díaz, S., & Herrero, A. 2014, *A&A*, 562, A135, doi: [10.1051/0004-6361/201322758](https://doi.org/10.1051/0004-6361/201322758)
- Sundqvist, J. O., & Owocki, S. P. 2013, *MNRAS*, 428, 1837, doi: [10.1093/mnras/sts165](https://doi.org/10.1093/mnras/sts165)
- Sundqvist, J. O., Petit, V., Owocki, S. P., et al. 2013, *Monthly Notices of the Royal Astronomical Society*, 433, 2497, doi: [10.1093/mnras/stt921](https://doi.org/10.1093/mnras/stt921)
- Timmes, F. X., & Swesty, F. D. 2000, *ApJS*, 126, 501, doi: [10.1086/313304](https://doi.org/10.1086/313304)
- Yusof, N., Hirschi, R., Meynet, G., et al. 2013, *MNRAS*, 433, 1114, doi: [10.1093/mnras/stt794](https://doi.org/10.1093/mnras/stt794)